Toolkit: Waves and Light

Waves

As astronomers, the information we get from the Universe comes to us almost exclusively in the form of light. Since almost everything we know about the cosmos is gleaned from the light we receive, we need to understand light and the way it interacts with matter in the most profound detail we can muster.

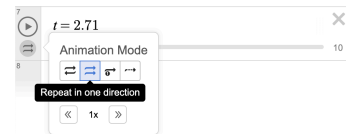
Light is a wave disturbance of the **electromagnetic field**. Freely-propagating electromagnetic waves are also known as **radiation**. To understand what these abstract concepts mean, we will first look at more tangible and familiar examples of waves. Some of the most familiar waves are surface waves of water. Later in life, we may notice waves on strings and we learn that sound is a pressure wave. All the simpler waves have something that they disturb as they **propagate**. This material is called the **medium**. Water waves disturb what would otherwise be a flat and horizontal surface between air and water. Waves on a stretched string disturb the straight resting position of the string. In matter, atoms and molecules are separated from one another by distances that are very close to their average separation. Sound waves are disturbances of these atoms and molecules away from these average separations.

Most physical waves are very complicated. For example, water waves near a beach change speed and shape as they move and often crest and crash into surf as they approach the shore. However, there are also very simple waves that are quite regular. The simplest waves undulate (crest and trough) in regular intervals in time called the **period** P of the wave. You can make such a wave in a bathtub by taking a ruler and bobbing it up and down regularly at the surface of the water. Straight wave fronts parallel to the ruler will move away from the ruler at a constant speed, and–provided you are bobbing at regular intervals–the distances between successive peaks will be the same for all successive peaks. This distance is called the **wavelength** λ of the wave.

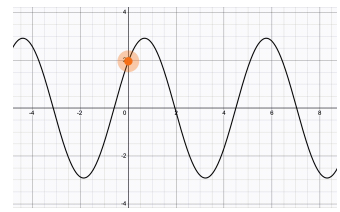
The reciprocal of the period $f = 1/P$ is called the **frequency** of the wave, but it is often more convenient to define the **angular frequency** $\omega = 2\pi f = \frac{2\pi}{P}$. The regular bobbing that you are doing with the ruler is called **simple harmonic motion**. An object undergoing simple harmonic motion is also said to be **oscillating** regularly. If there is a rubber duckie floating on the water then, when the wave reaches it, it will start to oscillate regularly. How much it bobs up and down is called the **amplitude** A of the wave. Note that the amplitude is independent of the wavelength: the wavelength is the distance between successive crests and the amplitude is the difference between the height of a crest and a trough.

1. Let's practice a bit with waves using Desmos.

a. Log into your account, so you can save your work. (If you don't have an account yet, please sign up for one.)
b. Make an orange dot that oscillates vertically around the origin with an adjustable period P and amplitude A. You will need to make sliders for time t, period P, and amplitude A. Adjust the ranges of the sliders to $t \in [0, 10]$, $P \in [0, \infty)$, and $A \in [0, 5]$. For time, you have to change the animation mode: click the arrows under the play button and select the option that jumps back to the beginning of the range when it gets to the end.



c. Make a sinusoidal graph $\sim \cos(x)$ that passes through the orange dot and moves to the right as the dot oscillates. The dot should track the point where the sinusoidal curve intersects the y-axis. To do this, you will have to remember from your math class how to transform functions. Your final answer will be a function of both x and t and contain the parameters A and P.



d. Modify your curve so that it has wavelength L by making a new slider for $L \in [0. \infty)$.
e. Manipulate L and P until you are able to deduce a formula for the speed with which the wave is moving to the right.

In a physical wave:

The speed v of a wave is determined solely by the medium/type of wave and not the wavelength or frequency.

For example, sound waves through air (at standard pressure and temperature) travel at around $330\ m/s$, but through water, they are about 4 times faster. Questions used to test whether you have understood this point will be on every test you will ever take about this material. They will try to trick you by saying things about wavelength and/or frequency, but that's a trap!

2. Use dimensional analysis to argue that the speed of surface waves in shallow water of depth h should be proportional to $\sqrt{gh}$. Can you use this to understand why waves crest and crash?

In problem 1e, you determined that the speed of the wave is related to the period and wavelength by $v = \lambda f$ where $f = 1/P$ is the frequency. Although there is nothing incorrect about your result, I find it misleading to leave it as written. As we have discussed, the speed v of the wave is set by the medium, not by the relation between wavelength and period. Instead, I prefer to write the formula in this way:

$$\boxed{\lambda = v\frac{1}{f}} \tag{1}$$

or as $f = v/\lambda$. This formula emphasizes that the wavelength of a wave is determined by its frequency (or frequency in terms of wavelength). Such a formula is called a *dispersion relation*.

To summarize, a simple wave $\psi(x,t)$ in one space dimension x and one time t can be put in the form

$$\psi(x,t) = A\cos(kx - \omega t) \tag{2}$$

This is a slight rewriting of your result in problem 1. In this expression:

- A is the amplitude.
- $\omega = 2\pi f = \frac{2\pi}{P}$ is the aforementioned angular frequency. The 2π is there because the periodicity of the cosine is 2π.
- Similarly, $k = \frac{2\pi}{\lambda}$ is called the (angular) wave number. Up to a factor of 2π, it is the reciprocal of the wavelength.

3. This problem is optional for students who have not yet taken calculus. (As usual, all students are encouraged to try this, but I won't make you learn derivatives and integrals from scratch.)
    a. Calculate the first and second derivatives $\partial\psi/\partial t$ and $\partial^2\psi/\partial t^2$ of the wave function with respect to time.[1]
    b. Calculate the first and second derivatives $\partial\psi/\partial x$ and $\partial^2\psi/\partial x^2$ of the wave function with respect to our single spatial direction.
    c. The combination $\Box := -\frac{1}{v^2}\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial x^2}$ is called the wave operator[2] and the equation $\Box\psi = 0$ is called the wave equation. Show that the function (2) satisfies the wave equation provided the dispersion relation (1) holds.

The spacetime in which we live consists of one time dimension and three spatial dimensions. It is not difficult to generalize the results above to accommodate this. The only major change is that the wave number k becomes a *wave vector* $\vec{k}$. As before, the length of this vector is the reciprocal of the wavelength (up to a factor of 2π) but now the direction of $\vec{k}$ can be interpreted as the direction of the wave.

4. Go back to your Desmos model and make a *new copy* of that work.
    a. Rewrite the new copy in terms of k and ω, but this time let k range also over negative values.
    b. What happens to the wave when you make k negative?

---

[1] If you have taken calculus, but are unfamiliar with this notation, don't forget to ask.

[2] The := notation means that the left-hand side (LHS) is being defined by the expression on the right-hand side (RHS). There are various different modifications of the "=" sign that people like to use to express this concept, but I like this one because ":" is read as "such that" in mathematics. Thus ":=" can be thought of as "such that it equals".

c. What is the analog of the wave described by equation (2) if we include also y- and z-directions?
d. How would you modify the dispersion relation (1) to accommodate these extra directions? (Hint: Review your vector math.)
e. If you have taken calculus, generalize problem #3 to one time and 3 spatial dimensions, and check that your 4D wave function satisfies the 4D wave equation, provided the wave vector and frequency satisfy the dispersion relation.

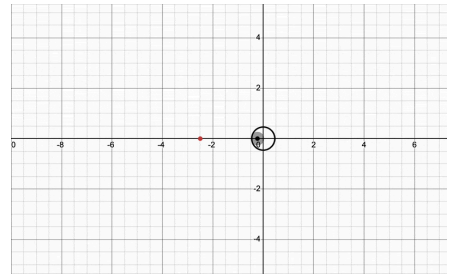
The Doppler effect

Let us model the waves emanating from a point source in two dimensions. An example of such a wave results from dropping a pebble in a calm pond.

5. Open up a new Desmos project and do the following:
   a. Make a slider for a time variable $t \in [0, 10]$ and set it to cycle back to 0 when it reaches 10 (just as you did in problem 1b).
   b. Make a second slider for a speed variable $v \in [0, 1.5]$.
   c. Make a dot that starts at the origin and moves with speed v to the left (*i.e.* along the negative x-axis).
   d. Set your speed slider to $v = 0$ and make a circular wave front that emanates from the origin and propagates outward as speed 1. You can do this by typing the equation $x^2 + y^2 = t^2$.
   e. Make another wave front that starts at time $t = 1$ and also emanates out at speed 1. You can do this with the following trick: on the line under that of part d, type out $x^2 + y^2 = sign(t - 1)\,(t - 1)^2$. Explain why this trick works.[3]
   f. We now want to tweak things so that the wave fronts are emitted from a moving point source. Set your speed slider to $v = 0.5$ and modify your wave front from part e so that it is emitted from the new position of your dot at time $t = 1$.
   g. Make a third wave front so that it is emitted at time $t = 2$ from the position $(-1, 0)$ where the source is at that time. Hint: Cut and paste your formula from part e and then modify it.
   h. We can keep cutting and pasting and modifying to make more wave fronts, but there is a more elegant way: Make a list of ten elements by writing $l = [0 \ldots 9]$. Turn off the wave fronts from parts d, e, and g, and write a new one that does all of these and more by writing $(x - l)^2 + y^2 = sign(t - l)(t - l)^2$. Explain why this works. (*I.e.* what is the $sign(t - l)$ for?)
   i. Move your speed slider around between $v = 0$ and $v = 0.9$. Describe what is happening to the wave fronts. Imagine our point source passing an observer sitting at the point $(-2, 0)$. What does this observer witness about the frequency as the emitter passes them?

[3] In other words, why does the new wave front appear only after 1 timestep?

In the problem above, we studied the properties of a wave with speed 1 emitted by an observer with variable speed v. You were supposed to notice that when you observe a wave emitted by something coming toward you, the apparent frequency of the wave is higher than it is when the emitter is at rest, because the wave fronts are bunched together. Similarly, the frequency is lower when the emitter is moving away from you, because the distance between crests (*i.e.* wavelength) is stretched out. This effect is called the **Doppler shift**.



6. Consider a source of sound moving *away from you* with speed v. Let $c > v$ denote the speed of the sound wave in air, and denote the true period of the source by P_0. The true frequency of the source is the reciprocal of this $f_0 = 1/P_0$. It is the frequency of the wave that the source is actually making and does not change if you move the source.
   a. In one true period P_0, how far has the source moved given that it is receding from you at speed v?
   b. The source emits a wave front every true period. Use this to argue that the distance between successive wave fronts according to you is $(c + v)P$. Draw a diagram depicting this situation. (Refer back to your Desmos work if that helps.)
   c. The distance between successive wave fronts according to you is the apparent wavelength λ, that is, the wavelength of the emitted wave *according to you*. Put these ingredients together to show that

$$\boxed{\lambda = \left(1 + \frac{v}{c}\right)\lambda_0} \tag{3}$$

   where λ_0 denotes the true wavelength of the wave the source is emitting.
   d. What is the equivalent formula for the frequency? Hint: recall the dispersion relation, but think carefully about which speed (v or c is actually relevant to this wave).
   e. Argue that when the source is moving toward you instead of away, then the result is simply to change $(1 + v/c) \rightarrow (1 - v/c)$. Thus, formula (3) and the one from 6d can be used with negative values of v, provided we interpret $v > 0$ as receding and $v < 0$ as approaching.

7. Go back to your Desmos simulation from problem 5. We are going to make a tone that indicates the shift.
   a. Add the line $tone(sign(\sqrt{5 - t})(1 + v)\,800)$. Then add another line $tone(sign(\sqrt{t - 5})(1 - v)\,800)$. You should be able to turn on sound in Desmos by clicking a button shaped like a speaker all the way at the top.

b. Mess around with these expressions until you understand what is happening. What is the factor of 800 for?
c. Put a red dot on the x-axis where the observer is who hears this Doppler shift as the source passes.
d. Optional: In your animation, the tone starts before the first wave front reaches the observer. Can you fix the code so that the tone does not start until the first wave front arrives?

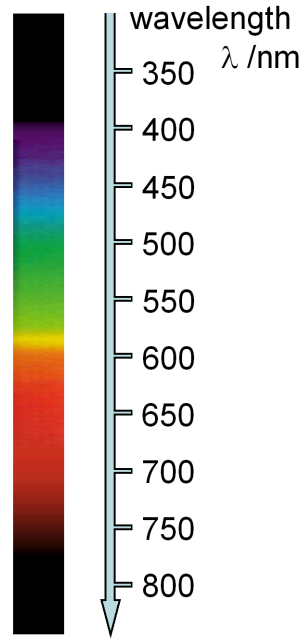
Intensity *vs.* pitch: amplitude *vs.* frequency

An effect that is often mistaken for the Doppler effect in the model above is the *intensity* (loudness) of the sound rather than the *pitch* (frequency): When an object that is making a sound is far away from us, the sound has a lower intensity than when it is closer to us. Suppose the noisy object is emitting sound in all directions evenly. This means that as the sound propagates outward, it is being spread evenly over a larger surface area. If the intensity of the emitter is constant at the source, that means that the intensity perceived as we move away from the source should fall off as $\propto \frac{1}{distance^2}$. We can refine our model so that it takes this new effect into account as well.

8. Continuing with your Desmos model from problem 7, we can add intensity to the tone function by putting a new value after the frequency separated by a comma.
   a. Add a line $tone(800, c)$ and make a slider for $c \in [0, 1]$. Turn off all the other tones (don't delete anything!) and play with the new slider until you are convinced that all that is happening is that the volume is increasing/decreasing.
   b. Make a tone function that increases/decreases as $\frac{1}{distance^2}$, and check that the intensity observed at your red dot symbolizing the observer is as it should be.
   c. Finally, do the following:
      i. Turn off all the tone functions.
      ii. Copy your old tone functions from problem 7 (without deleting them), and copy them onto new lines.
      iii. Modify your tone functions so that the volume increases/decreases as it should in addition to having the correct Doppler shift.
      iv. Variously switch on/off the intensity/Doppler (e.g. by holding the emitter still) until you have convinced yourself that you understand the differences between these effects.

In our derivation of the Doppler formula, we used sound waves as an example because they are more familiar to us, but our derivation was independent of the type of wave. Since light is a wave as well, we might ask whether it too suffers a Doppler shift if there is relative motion between the observer and the source. This is one of the most important ideas in observational astronomy: if we know what type of light something is emitting, then comparing the wavelengths that we *observe* to what they *should* be, we can determine the component of the velocity of the source directed toward/away from us.

9. An astronomical source is known to emit light with a wavelength of $\lambda_0 = 560\ nm$.
    a. Is this visible light? If so, what color is it?
    b. When we observe this object, we find instead that $\lambda = 460\ nm$.
        i. What color does this look like to us?
        ii. Is this object moving toward us or away from us?
        iii. What is the speed of the source? What fraction of the speed of light (in vacuum) is this?



In the problem above, the light's wavelength is shifted to the shorter side. In the visible part of the spectrum, blue light is on the short side (450 – 485 nm) whereas red light is on the longer side (625 – 750 nm). Because of this, we refer to light that is Doppler shifted to shorter apparent wavelengths as **blue-shifted** and light that is Doppler shifted to longer apparent wavelengths as **red-shifted**. We use this terminology not just for light, but for other waves as well.

Note that the Doppler shift we have been discussing can only tell us about the part of the relative motion that is directly toward/away from us. This type of motion is called **radial motion**. (The transversal component of the motion is called **tangential motion** (because this component is tangent to the celestial sphere), but the Doppler shift as we have discussed it, does not tell us anything about that part.)

10. The visible matter in the Universe is almost entirely made up of hydrogen. As we will learn in our next toolkit, hydrogen atoms emit light with a wavelength $\lambda_0 = 656.3\ nm$ called the hydrogen-alpha line or $H\alpha$ for short. Suppose we observe a particular galaxy's $H\alpha$ line to have an apparent wavelength of $\lambda_0 = 660.0\ nm$. Describe the radial motion of this galaxy quantitatively.

Optics

Simple waves are often described by the geometry of their wavefronts. The wavefronts in your 2D Desmos simulation are circular, so we might describe these as circular waves. In 3D, however, the analogous wave would be spherical, so these are usually called spherical waves. The simplest geometries are planar, so we might consider making plane waves. In a 2D simulation, the wavefronts would look like parallel lines.

11. Make a new copy of your Desmos model of problem 5 and modify it so that you get plane waves with vertical wavefronts starting at the point $x = -10$ and moving to the right.

Above, we emphasized that the speed of a wave depends not on the wavelength or frequency but on the *medium* through which the wave propagates. For example, sound waves in air are about 4 times slower than they are in water.

An of-repeated statement we have all heard is that the speed of light is fixed to be

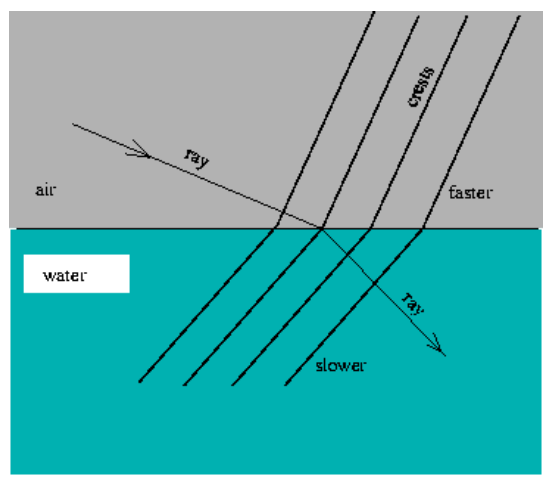
$$c = 2.99792458 \times 10^{8}\, m/s \tag{4}$$

The correct statement is that any inertial observer who measures the speed of light *in vacuum* will find it to have this magnitude. When we say "nothing can go faster than the speed of light", we really mean that no information can be transferred faster than the speed of light in vacuum. When light passes through transparent media like water or glass, it slows down. Other particles can travel faster through these media (as long as their speed does not exceed c), but this will produce shock waves of light called Cherenkov radiation.

For an optically transparent medium, we define the **index of refraction**: instead of working directly with the speed of light in that medium $v \leq c$, we work with the ratio

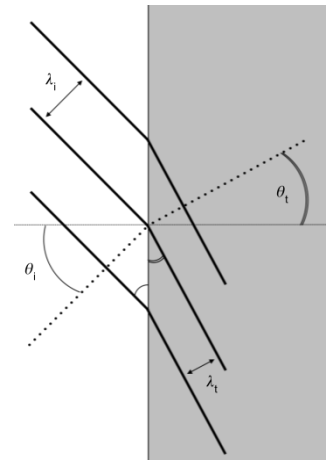
$$n = \frac{c}{v} \in [1, \infty) \tag{5}$$

of the speed of light in vacuum to the speed in the medium. Note that $n = 1$ only in free space and is larger than that for any other transparent medium. The strange name comes from one of the hallmarks of waves: they **refract**. Refraction happens when a wave is incident on a medium that is transparent to that wave. As the wave passes into the medium, it changes speed because the medium sets the speed of the wave. If the wave slows down, by equation (1), its wavelength will get smaller. If the wavelength is incident on a medium/medium boundary at some angle, continuity of the wavefronts will force the wave to turn, as indicated in the figure at right.



12. In this Desmos demo, the right half-plane is filled with a transparent medium. On the LHS the index of refraction is 1, and on the shaded RHS it is $n > 1$. The interface can be tilted by changing the angle α between the plane and the horizontal axis. The demo can be animated by clicking the play button for a. The default settings are such that $N = 20$ wavefronts are created at $x_0 = -5$ units to the left of the interface as a progresses, after which the animation starts over.
    a. Start the animation in the default setting and describe what happens to the wave fronts when they enter the medium.
    b. Pause the animation, and slide α so that the fonts hit the interface obliquely. Notice that individual wave fronts stay connected across the interface. Describe

in your own words why this and the change in speed make the plane wave change direction.



The figure at right shows a close-up of wavefronts with wavelength λ_i incident on the interface at an angle θ_i. In the material, the same wave has wavelength λ_t and the fronts form an angle θ_t with the interface.

c. The angle of the fronts with the interface is equal to the angle of the normals of the fronts with the normal to the interface. Use this to show that $\frac{\sin\theta_i}{\lambda_i} = \frac{\sin\theta_t}{\lambda_t}$.

d. Continuity of the wave fronts across the boundary also implies that the frequency of the wave cannot change across the interface. Use this, the dispersion relation (eq 1), and the definition of the index of refraction to prove **Snell's Law of Refraction**:

$$n_i \sin\theta_i = n_t \sin\theta_t \tag{6}$$

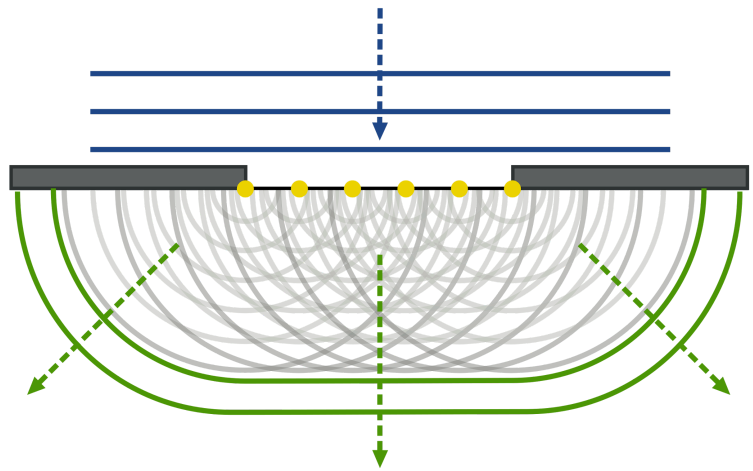
(This law was used to make the Desmos demo: If you look in the "stuffs" folder, it is used to define the angle β.)

Refraction is an important property of waves that has no analog for particles. Other important such properties are **diffraction** and **interference**. Similarly to reflection and refraction, diffraction is a type of change in the direction of motion as a wave encounters an obstacle with an associated size that is small relative to the wavelength of the wave. There are two main cases: The first is that the obstacle is a small object like a buoy or a piling is to an ocean wave. In this case the wave easily bends around the obstacle. In the second case, the obstacle is very large but there is an opening with a size small or comparable to the wavelength. When this happens, the wave "fans out" when it passes through the opening, as in the picture to the right.

The specific patterns that you get from waves interacting with obstacles can be understood from Huygens' Principle, which states that for a propagating wave, the next wavefront will have the same shape as you would get from treating each point on the wave front as a source of spherical waves and taking the next wavefront to be the "envelope" of the

spherical wavefronts. The schematic diagram below depicts this for the wave in the picture above. Note that as the opening becomes smaller, the resulting envelopes look more spherical.

Finally, interference is the phenomenon that two waves of the same type will change each other's shape when they encounter each other. For simple waves, as we have been discussing here, the interference of two waves is gotten by simply adding them together. For waves that displace a medium like water or sound waves, this just means that we sum up the displacements. Since displacements can be both positive and negative, this means that the effect is sometimes to amplify the wave and other times to (partially) cancel out. We say that the waves are interfering **constructively** or **destructively**, respectively.

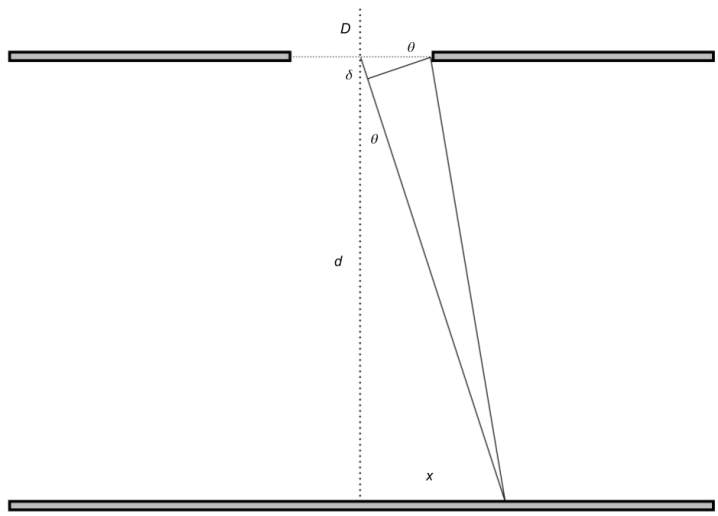


13. Go back to your Desmos project from problems 1 and 4.
    a. Make a copy and change things around so that you have two waves $f_1(x,t)$ and $f_2(x,t)$ each with their own (adjustable) velocities $v_{1,2} \in [-1, 1]$ amplitudes $A_{1,2} \in [0, 1]$, and wave numbers $k_{1,2} \in [0, 10]$. Make a time slider with $t \in [0, 50]$ and adjust it so that it cycles through instead of bouncing back and forth when you animate it.
    b. Make a third wave $f(x,t) = 3\left[f_1(x,t) + f_2(x,t)\right]$ by adding them together. We are multiplying it by 3 to make the resulting wave stand out from the component waves (but feel free to remove this factor). Play with the various slider settings and see if you can do anything interesting. Of course, you should make a note of what is happening.
    c. Set $A_1 = A_2$, $k_1 = k_2$, and $v_1 = -v_2$ so that the two waves are the same but traveling in opposite directions. Animate the time variable and describe what you are seeing.
    d. Now set $v_2 = 2v_1$ and describe the resulting pattern.

The pattern you found in 13c. is called a **standing wave**. The points where the waves cancel are called **nodes**. This is where the two waves are experiencing complete destructive interference. The opposite points where they are experiencing complete constructive interference are called **antinodes**. These points are sitting still in part c, but as you saw in part d, you can make them move by changing the relative velocities.

We now combine diffraction and interference to describe an important phenomenon that limits the ability of telescopes to resolve objects. When light of wavelength λ comes into a

telescope, it either passes through a lens (refractive telescope) or it is bounced off a mirror (reflective scope) called the **objective**. Suppose this objective has diameter D. In either case, the light of wavelength λ must pass through a "hole" of size D. When this happens, we know from our work above that there must be some diffraction. In the next two problems, we apply Huygens' Principle to see what happens to light as it passes through an opening.



14. Refer to the diagram at right. It is intended to display an aperture of size D a distance d away from a screen. A point is indicated on the screen a distance x from the middle (which is indicated by the vertical dotted line). The distance from x to the middle of the slit differs from that to the near side of the slit by an amount δ as indicated in the figure.
    a. Argue that the two angles denoted by θ are the same.
    b. Using the implied right triangle, write an expression relating δ, D, and θ.
    c. Argue that the waves arriving from the middle and right edge of the opening will interfere at x maximally constructively if $\delta = n\lambda$ for integer $n \in \mathbf{Z}$ and maximally destructively if $\delta = (n + \frac{1}{2})\lambda$. The former waves are said to arrive *in phase* and the latter *out of phase*.
    d. On a calculator or Desmos, calculate how large an angle θ can be before θ exceeds $\sin\theta$ by 1% or more. You should do this in radians first and only convert to degrees at the end. Use this "small-angle formula" $\sin\theta \approx \theta$ to show that the first minimum occurs at an angle of

$$\theta \approx \frac{\lambda}{D} \quad \text{(radians)} \tag{7}$$

    e. On a calculator or Desmos, calculate how large an angle θ can be before $\tan\theta$ exceeds $\sin\theta$ by 1% or more. You should do this in radians first and only convert to degrees at the end.
    f. Use the small-angle approximation $\tan\theta \approx \sin\theta$ to show that the distance from the center to the first minimum is $\frac{\lambda}{D}d$.

In the problem above, we compared the middle of the slit to the right edge because then, for any point between the two, there is a corresponding point in the left half. Thus, by symmetry, waves emanating from points in the right half will interfere the same way as just described with the corresponding points in the left half. According to Huygens, we should add up the interference

of all these points to find that the light passing through the aperture interferes with itself in such a way as to make a pattern of alternating light and dark spots.

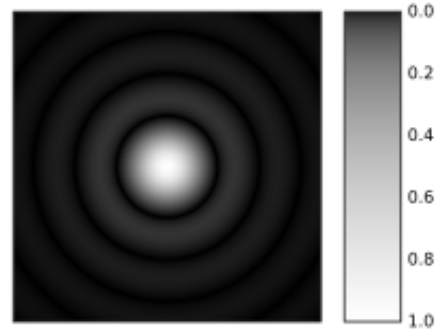
15. Using Desmos, we can explicitly model the pattern that results from Huygens' Principle. We already saw in the previous problem that we need to add waves that are arriving at points on the screen in different phases. Suppose we add two identical sine waves that are slightly out of phase. Then we will need to add sine waves with different arguments. In Math 4, we (will) learn that $\sin(a) + \sin(b) = 2\cos\left(\frac{a-b}{2}\right)\sin\left(\frac{a+b}{2}\right)$.
    a. Use this to show that a wave $A\sin(\omega t)$ adds to a phase-shifted wave with the same frequency and amplitude $A\sin(\omega t + \varphi)$ to give a wave of frequency ω and amplitude $A\cos\left(\frac{\varphi}{2}\right)$.

    If, following Huygens, we split the aperture into N identical point sources all emitting spherical waves of frequency ω, we will need to add N sine waves all with phases shifted by a phase φ. Let's do this on Desmos for a variable integer N and plot the result.
    b. Open Desmos and make a new project called "Diffraction" or something like that. Type $N = floor(M)$, and make a slider for $M \in [1, 100]$. Slide M around and look at what happening to the value of N. Explain what the floor function does.
    c. Set $M = 1$, and make a list of the first N natural numbers by typing $l = [1 \ldots N]$. Make a new one centered around 0 by defining $L = \frac{N}{2} - l$.
    d. Plot a list of cosines by typing $\cos(Lx)$ and vary N using the slider for M. These are the cosines with different phases that Huygens is telling us to add together. Long-click on the color of the lines to open up the options box, and make the line thickness 0.5. This will make everything look less cluttered.
    e. Following Huygens, add up all the cosines by typing $\frac{1}{N}\sum_{n=1}^{N}\cos\left(\left(\frac{N}{2} - n\right)x\right)$. (Hint: type "sum" to make the $\sum$ symbol and then fill in the rest.[4]) We are adding the factor of $\frac{1}{N}$ because we are splitting the wave into N parts and adding them back together. (We should not get a wave that has N times the original amplitude!) Vary N again and write down what you observe.
    f. We actually want the intensity of the wave, which is given by squaring the amplitude, so square the function in part e and plot that. Again, vary the value of N and report what you see.
    g. We learned in and around problem 8 that the intensity of light depends on the distance from the source. As the magnitude of x in problem 15 increases, the intensity of the light arriving at x decreases like $\sim\frac{1}{|x|^2}$. In this case, the source is

[4] This symbol is a larger version of the uppercase Greek letter "Sigma". It is used in mathematics as an abbreviation for "summation": adding a bunch of copies of the argument with different values of a counting number that ranges from the indicated minimum to the indicated maximum.

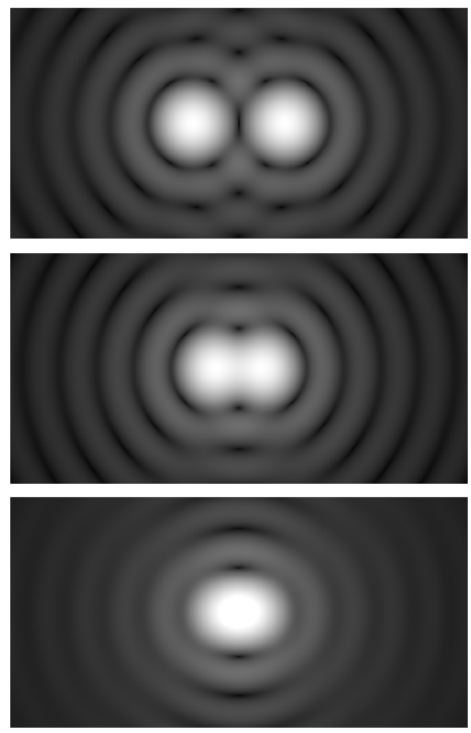
never closer to the screen than the distance d from problem 15. Model this by plotting the function $\frac{1}{1+x^2}$ instead.

h. Finally, incorporate the effect from part g by multiplying your answer in part f by $\frac{1}{1+x^2}$. Describe the resulting diffraction pattern. The image at right is called the Airy diffraction pattern. How does your result compare to this Airy disk?



Phew! That was a lot. In the last few problems, we worked hard to arrive at the conclusion that for a wave of wavelength λ passing through an opening of size D, we can expect diffraction of the wave of angular size set by the ratio $\frac{\lambda}{D}$ as you derived in equation (7).

Now suppose we are looking through a telescope of aperture size D at two objects in the sky with an angular separation α. According to problem 15, the light from each object will make an Airy diffraction pattern, so we see something like one of the cases at right. It becomes clear that once the central disks in the Airy patterns start to merge, we will lose the ability to distinguish a single source from a double source. Determining the **diffraction-limited resolution** of a telescope in this way is called the *Rayleigh criterion*. By definition, this is a limit on the resolving power of the telescope, and it depends only on the wavelength of the light as it compares to the aperture size of the telescope.



Full disclosure: The pattern in problem 16 is not *exactly* (a cross section of) the Airy disk, but it is very close. Reanalyzing the case of a circular 2D aperture (rather than the 1D from problem 16) results in the true Airy pattern which has slightly displaced maxima/minima. For example, the Rayleigh criterion reported here is $\theta \approx 1.22\,\frac{\lambda}{D}$.

We close by emphasizing that this diffraction limitation is the best-case scenario, not a problem that can be overcome with better technology (of the same aperture size). Any real telescope will *at best*, be limited by this fundamental behavior of waves as opposed to, for example, limits on the precision of the manufacturing process. To express this, we say that a telescope is **diffraction-limited** to a certain resolution.